# Jonas Elias El Gammal

Cosmologist | Scientific machine-learning researcher | Scientific software developer

jonas.el.gammal@rwth-aachen.de | Tallinn, Estonia | jonaselgammal.github.io | GitHub | ORCID

## PROFILE

Postdoctoral researcher working across cosmology, gravitational-wave astronomy, Bayesian inference, probabilistic machine learning, and computational statistics. I develop open-source methods that turn expensive physical models into tractable, validated inference workflows.

## RESEARCH EXPERIENCE

**Postdoctoral Researcher - KBFI / NICPB, Tallinn** 2026-present
- Research in cosmology, gravitational-wave inference, scientific machine learning, and reusable computational methodology.

**Postdoctoral Researcher - University of Insubria** 2025-2026
- Developed GPU-accelerated and differentiable pulsar-timing-array inference with JAX in the EPTA collaboration.

**Visiting Researcher - CERN** 2024
- Developed a LISA analysis pipeline for scalar-induced gravitational waves.

## EDUCATION

**PhD in Physics - University of Stavanger** 2021-2025
Developed data-analysis methods for LISA, including Gaussian-process-accelerated Bayesian inference and reconstruction of primordial curvature perturbations from scalar-induced gravitational waves.

**Erasmus Exchange - Grenoble INP, PHELMA** 2019-2020
Specialisation in nuclear engineering.

**MSc in Physics - RWTH Aachen University** 2018-2020
Grade 1.3 (approximately 3.7/4.0 GPA). Thesis: *Accelerating Bayesian Inference of Expensive Likelihoods with Gaussian Processes*.

**BSc in Physics - RWTH Aachen University** 2015-2018
Grade 1.7 (approximately 3.3/4.0 GPA). Focus on astroparticle physics and cosmology.

## TEACHING

**Numerical Methods**, University of Stavanger - full bachelor-level course (2023, 2024): Python, numerical differentiation and integration, ODEs, and PDEs.

**Electromagnetism and Special Relativity**, University of Stavanger - full bachelor-level course (2023).

**Physics and data-processing tutorials**, RWTH Aachen University (2017-2019).

## METHODS AND TECHNOLOGY

**Research:** Bayesian inference, Gaussian processes, active learning, uncertainty quantification, Monte Carlo methods, numerical optimisation, stochastic simulation, time-series analysis, cosmology, and gravitational-wave data analysis.

**AI systems:** LLM agents, tool use, agent loops, multi-agent workflows, typed planning, structured outputs, validation, repair, tracing, evaluation, and reproducible experimentation.

**Software:** Python, JAX, NumPy, SciPy, scikit-learn, PyTorch, TensorFlow, C++, Git, Docker, Linux, SLURM, HPC/GPU computing, CI/CD, testing, documentation, and package development.

## LANGUAGES

**German** - native | **English** - near-native | **French** - B2/C1

## PUBLICATIONS

**Are PTA measurements sensitive to gravitational wave non-Gaussianities?**
C. Cecchini, J. El Gammal, G. Franciolini, M. Pieroni - 2026 - under review; arXiv:2605.05157

**Accelerating LISA inference with Gaussian processes**
J. El Gammal, R. Buscicchio, G. Nardini, J. Torrado - 2025 - Physical Review D 112, 063010

**Reconstructing primordial curvature perturbations via scalar-induced gravitational waves with LISA**
J. El Gammal et al. (LISA Cosmology Working Group) - 2025 - JCAP 05, 062

**Parallelized Acquisition for Active Learning using Monte Carlo Sampling**
J. Torrado, N. Schoeneberg, J. El Gammal - 2023 - arXiv:2305.19267

**Fast and robust Bayesian inference using Gaussian processes with GPry**
J. El Gammal, N. Schoeneberg, J. Torrado, C. Fidler - 2023 - JCAP 10, 021

## CURRENT RESEARCH AND DEVELOPMENT

- High-dimensional GP surrogate inference using gradient-based NUTS acquisition, with emphasis on numerical robustness, reproducible benchmarking, and scalable active learning.
- GPU-accelerated JAX likelihoods and reversible-jump inference for continuous gravitational-wave searches with pulsar timing arrays.
- PROSECCO: differentiable surrogate refitting of physical stochastic-background models against model-independent LISA reconstructions (in development).
- Evaluation environments for autonomous scientific agents that propose, test, and revise executable structural theories under noisy and incomplete data (early research).

## OPEN-SOURCE SOFTWARE

**GPry** - Gaussian-process surrogate modelling and active learning for Bayesian inference with expensive likelihoods.

**SIGWAY** - JAX framework for scalar-induced gravitational-wave simulation and inference.

**discoverysamplers** - Consistent interfaces connecting Discovery PTA models to multiple Bayesian inference backends.

**fastropop** - JAX-backed simulation of supermassive black-hole-binary populations and their nanohertz signals.

**Graphsmith** - Typed graph planning with deterministic compilation, validation, local repair, and traceable execution.